

July2025 CSE330

Online 1 (B1/B2)

Time: 40 min

You need to implement:

Complete the code sections that have '# TODO' in the comments.

- Windowing Average Imputation:

For a window of size k ,

$$x_{i,j}^{\text{new}} = \begin{cases} x_{i,j}, & x_{i,j} \neq \text{NaN} \\ \frac{1}{k} \sum_{l=i-\lfloor k/2 \rfloor}^{i+\lfloor k/2 \rfloor} x_{l,j}, & x_{i,j} = \text{NaN} \text{ and full window available} \\ \text{Missing values can be ignored,} & \text{if full window is not available (start/end of data)} \end{cases}$$

- Tanh:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

- Tanh Gradient:

$$\frac{d \tanh(x)}{dx} = 1 - \tanh^2(x)$$

gradient:

$$\frac{\partial L}{\partial x} = dout \cdot (1 - \tanh^2(x))$$

- MAE:

$$\text{MAE}(y_{\text{pred}}, y_{\text{true}}) = \frac{1}{n} \sum_{i=1}^n |y_{\text{pred},i} - y_{\text{true},i}|$$

- MAE Gradient:

$$\frac{\partial |y_{\text{pred}} - y_{\text{true}}|}{\partial y_{\text{pred}}} = \text{sign}(y_{\text{pred}} - y_{\text{true}})$$

$$\frac{\partial L}{\partial y_{\text{pred},i}} = \frac{1}{n} \text{sign}(y_{\text{pred},i} - y_{\text{true},i})$$

$$\text{sign}(z) = \begin{cases} +1, & z > 0 \\ 0, & z = 0 \\ -1, & z < 0 \end{cases}$$

- Inference of test data:

Load test data and evaluate MAE.

Caution:

1. You can only use NumPy and Pandas (import statements are already present in the skeleton code; you cannot add or remove any import statements)
2. You should compute using vectorized NumPy operations. Any loop will result in some penalty, except for Windowing Average Imputation.

Output to be matched:

Test MAE: 0.526722